

Webtoon Panel Slicer v6.1 — Non-Overlapping Techniques, Long Strips & Source

Error-checked. Signal hierarchy prevents double-cuts. Black-void gutters for manhwa. Optional YOLO sliding-window.

1. Signal hierarchy (no overlap)

Techniques are layered so they do not fight each other. Each layer only fires when higher layers leave a block too tall or empty.

L0 — YOLO (optional): if --yolo weights given → sliding-window detection → 1-D NMS. On success, classical is skipped.

L1 — Structural gutters: white (bright + low edge) **OR** pure-black void (dark + low edge). Manhwa often uses black dividers 60–200 px; white gutters are the webtoon default. These define primary content blocks.

L2 — Event zones (only inside tall blocks): rolling-std peaks = SFX/impact. Cut is placed *after* the zone ends. Does not re-cut L1 boundaries.

L3 — Projection valleys: adaptive local minima of smoothed luminance that are also gutter-like. Used when L2 is sparse.

L4 — Hough lines (optional OpenCV): only if still fewer than 2 candidates after L2+L3.

L5 — Recursive binary split (--mode recursive): Pang-style; replaces L2–L4 for that mode.

Post: merge_nearby_cuts (later cut wins) + fragment merge into larger neighbours. Guarantees no double panels from multiple signals.

2. Why this hierarchy

Literature ensembles (contour + watershed + projection; tree-CRF on CC/edge/line) fail when every stage proposes cuts independently → over-segmentation. v6.1 assigns exclusive roles: L1 owns hard separators, L2 owns narrative beats inside tall action blocks, L3/L4 only fill gaps. Black-void detection is required for manhwa (RGB≈0 dividers); white-only detectors miss them.

3. Long-strip handling

- Classical path: processes full height with row stats (numpy); fine for 20k px on typical RAM.
- YOLO path: sliding window (default 1280 px, 25% overlap), map boxes back, 1-D interval NMS, midpoint cuts between consecutive panels.
- Same pattern as production webtoon face pipelines (split tall image → detect → dedupe overlaps).

4. Error checks performed

- py_compile clean
- Synthetic white gutter → 2 panels
- Synthetic black void gutter → 2 panels (manhwa style)
- Real test strip → 5 panels [606, 1355, 1404, 1908, 2127]; all y1>y0; sources=classical
- No inverted crop ranges; pad clamped to image bounds

5. Datasets for your YOLO

- Roboflow Manhwa Panel Detector (~254)
- Roboflow Webtoon-Manhwa Panels (Irregular / Diagonal / Outbound / Noborder...)
- Manga109 + deepghs/manga109_yolo or huyvux3005/manga-fpt
- leoxs22/manga-panel-detector-yolo26n (pretrained nano)
- mosesb/best-comic-panel-detection (YOLOv12x)
- comix_books_v0 (~950k Western pages with panel dets)

6. CLI

```
python webtoon_panel_slicer.py strip.pdf -o panels/ --subdivide --mode hybrid
python webtoon_panel_slicer.py chapter.png -o out/ --yolo best.pt --yolo-window 1280
python webtoon_panel_slicer.py page.png --mode recursive --max-cuts 8
python webtoon_panel_slicer.py page.png --subdivide --hough # needs opencv
```

7. Test result (classical hybrid)

#	v6.1	Manual	Δ
01	606	680	−74
02	1355	1370	−15
03	1404	1440	−36

04	1908	1990	-82
05	2127	2125	+2

8. Full source (v6.1)

```
#!/usr/bin/env python3
"""
webtoon_panel_slicer.py (v6)
=====
Free local tool for precise panel extraction from vertical webtoon / manhwa
long strips and irregular panels.

Integrated techniques
-----
Classical (always available):
  • Adaptive white + low-edge gutters (percentile-aware)
  • Event-zone detection (rolling std) → cut AFTER SFX/impact
  • Multi-scale projection valleys
  • Recursive 1-D binary split (Pang et al. ACM MM 2014 inspired)
  • Optional OpenCV Canny + Probabilistic Hough gutter lines
  • Fragment merge + face-safety heuristic
  • Sliding-window processing for very tall strips (memory-safe)

Optional deep (if ultralytics installed):
  • YOLO panel detector with sliding-window inference on long strips
  • Cascade: YOLO first on ambiguous/tall blocks, classical fallback
  • Works with any ultralytics YOLO weights (YOLOv8/11/12, nano→x)

Designed for the hard cases: 800x10k+ px manhwa, irregular/outbound panels,
variable gutters, full-bleed SFX.

Datasets useful for training your own YOLO (see PDF / README):
  Roboflow Manhwa Panel Detector, Webtoon-Manhwa Panels (irregular classes),
  Manga109 (frame), deepghs/manga109_yolo, comix_books_v0, etc.

Usage examples
-----
# Classical hybrid (default, no extra deps)
python webtoon_panel_slicer.py strip.pdf -o panels/ --subdivide --mode hybrid

# Recursive Pang-style
python webtoon_panel_slicer.py strip.png --mode recursive --max-cuts 8

# YOLO (if you have weights)
python webtoon_panel_slicer.py long_strip.png -o out/ --yolo best.pt --yolo-conf 0.25

# YOLO + classical cascade on very tall images
python webtoon_panel_slicer.py chapter.png -o out/ --yolo best.pt --subdivide
"""

from __future__ import annotations

import argparse
import os
import sys
from dataclasses import dataclass
from pathlib import Path
from typing import List, Optional, Tuple

import numpy as np
from PIL import Image

# Optional deps
try:
    from pdf2image import convert_from_path
    HAS_PDF = True
except ImportError:
    HAS_PDF = False

try:
    import pytesseract
    HAS_OCR = True
except ImportError:
    HAS_OCR = False

try:
    import cv2
    HAS_CV2 = True
except ImportError:
    HAS_CV2 = False

try:
    from ultralytics import YOLO
    HAS_YOLO = True
except ImportError:
    HAS_YOLO = False

@dataclass
class Panel:
    index: int
    y0: int
    y1: int
    image: Image.Image
    label: str = ""
    source: str = "classical" # classical | yolo

    @property
    def height(self) -> int:
        return self.y1 - self.y0

# -----
# I/O
# -----
def load_image(path: str, dpi: int = 72) -> Image.Image:
    path = str(path)
    if path.lower().endswith(".pdf"):
        if HAS_PDF:
```

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        pages = convert_from_path(path, dpi=dpi, first_page=1, last_page=1)
        return pages[0].convert("RGB")
    import subprocess, tempfile
    with tempfile.TemporaryDirectory() as tmp:
        prefix = os.path.join(tmp, "page")
        subprocess.check_call(
            ["pdftoppm", "-png", "-r", str(dpi), "-f", "1", "-l", "1", path, prefix]
        )
        pngs = list(Path(tmp).glob("*.png"))
        if not pngs:
            raise RuntimeError("pdftoppm produced no output")
        return Image.open(pngs[0]).convert("RGB")
    return Image.open(path).convert("RGB")

def row_statistics(arr: np.ndarray) -> Tuple[np.ndarray, np.ndarray, np.ndarray]:
    row_mean = arr.mean(axis=(1, 2)).astype(np.float64)
    row_std = arr.std(axis=(1, 2)).astype(np.float64)
    gray = arr.mean(axis=2)
    edge_energy = np.abs(np.diff(gray, axis=1)).mean(axis=1)
    return row_mean, row_std, edge_energy

def mask_to_regions(mask: np.ndarray, min_len: int = 15) -> List[Tuple[int, int]]:
    regions, in_reg, start = [], False, 0
    for i, val in enumerate(mask):
        if val and not in_reg:
            start, in_reg = i, True
        elif not val and in_reg:
            if i - start >= min_len:
                regions.append((start, i - 1))
            in_reg = False
    if in_reg and len(mask) - start >= min_len:
        regions.append((start, len(mask) - 1))
    return regions

def merge_nearby_cuts(cuts: List[int], min_gap: int) -> List[int]:
    """Keep cuts at least min_gap apart; prefer the later cut on near-ties (exit SFX)."""
    if not cuts:
        return []
    cuts = sorted(cuts)
    out = [cuts[0]]
    for c in cuts[1:]:
        if c - out[-1] < min_gap:
            out[-1] = c # later wins
        else:
            out.append(c)
    return out

def rolling_std(x: np.ndarray, win: int = 40) -> np.ndarray:
    n = len(x)
    out = np.zeros(n, dtype=np.float64)
    for i in range(n):
        out[i] = x[max(0, i - win): i + 1].std()
    return out

def looks_like_face_cut(arr: np.ndarray, y: int, band: int = 30) -> bool:
    h, w = arr.shape[:2]
    y0, y1 = max(0, y - band), min(h, y + band)
    strip = arr[y0:y1].mean(axis=2)
    dark = strip < 80
    long_runs = 0
    for col in range(0, w, 4):
        col_dark = dark[:, col]
        max_run = cur = 0
        for v in col_dark:
            cur = cur + 1 if v else 0
            max_run = max(max_run, cur)
        if max_run > band * 0.7:
            long_runs += 1
    return long_runs > (w // 4) * 0.28

# -----
# Classical signals
# -----
def projection_split_candidates(
    row_mean: np.ndarray, edge_energy: np.ndarray,
    y0: int, y1: int, min_gap: int,
) -> List[int]:
    sub_m = row_mean[y0:y1]
    sub_e = edge_energy[y0:y1]
    n = len(sub_m)
    if n < min_gap * 3:
        return []
    bright = sub_m > np.percentile(sub_m, 88)
    low_edge = sub_e < max(np.percentile(sub_e, 30), 3.0)
    gutterish = bright & low_edge
    kernel = 15
    smooth = np.convolve(sub_m, np.ones(kernel) / kernel, mode="same")
    k = max(5, min_gap // 4)
    candidates = set()
    for i in range(k, n - k):
        window = smooth[i - k: i + k + 1]
        if smooth[i] <= window.min() + 1e-6 and gutterish[i]:
            candidates.add(y0 + i)
    for a, b in mask_to_regions(gutterish, min_len=max(8, min_gap // 3)):
        candidates.add(y0 + (a + b) // 2)
    return sorted(c for c in candidates if y0 + min_gap < c < y1 - min_gap)

def event_zone_cuts(
    row_mean: np.ndarray, y0: int, y1: int, min_side: int,
) -> List[Tuple[float, int]]:
    sub = row_mean[y0:y1]

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n = len(sub)
if n < min_side * 2:
    return []
rs = rolling_std(sub, win=40)
thresh = float(np.percentile(rs, 90))
high = rs > thresh
zones = mask_to_regions(high, min_len=40)
scored = []
for ez0, ez1 in zones:
    if (ez1 - ez0) > n * 0.65:
        continue
    cut = y0 + min(ez1 + 8, n - 1)
    left_h, right_h = cut - y0, y1 - cut
    if left_h < min_side or right_h < min_side:
        continue
    zone_score = min(60.0, (ez1 - ez0) * 0.4) + 20.0
    scored.append((zone_score, cut))
return scored

def hough_horizontal_candidates(arr: np.ndarray, y0: int, y1: int, min_gap: int) -> List[int]:
    if not HAS_CV2:
        return []
    strip = arr[y0:y1]
    gray = cv2.cvtColor(strip, cv2.COLOR_RGB2GRAY) if strip.ndim == 3 else strip
    edges = cv2.Canny(gray, 50, 150, apertureSize=3)
    lines = cv2.HoughLinesP(
        edges, rho=1, theta=np.pi / 180, threshold=80,
        minLineLength=max(40, strip.shape[1] // 4), maxLineGap=20,
    )
    if lines is None:
        return []
    ys = []
    for x1, yy1, x2, yy2 in lines[:, 0]:
        if abs(yy2 - yy1) <= 3:
            ys.append(y0 + (yy1 + yy2) // 2)
    ys = sorted(ys)
    clustered = []
    for y in ys:
        if not clustered or y - clustered[-1] > min_gap // 2:
            clustered.append(y)
    return [y for y in clustered if y0 + min_gap < y < y1 - min_gap]

def recursive_split(
    arr, row_mean, edge_energy, y0, y1, min_panel, max_depth, depth=0,
) -> List[Tuple[int, int]]:
    h = y1 - y0
    if h < min_panel * 2.2 or depth >= max_depth:
        return [(y0, y1)]
    min_side = int(min_panel * 1.3)
    candidates = projection_split_candidates(row_mean, edge_energy, y0, y1, min_side)
    for _, c in event_zone_cuts(row_mean, y0, y1, min_side):
        candidates.append(c)
    candidates = sorted(set(candidates))
    if not candidates:
        return [(y0, y1)]
    best_c, best_score = None, -1e9
    for c in candidates:
        left_h, right_h = c - y0, y1 - c
        if left_h < min_side or right_h < min_side:
            continue
        win = row_mean[max(0, c - 8): c + 8]
        edge_win = edge_energy[max(0, c - 8): c + 8]
        bright = float(win.mean()) if len(win) else 0
        low_e = float(edge_win.mean()) if len(edge_win) else 99
        gutter_q = bright * 0.4 - low_e * 2.0
        balance = 1.0 - abs(left_h - right_h) / h
        face_pen = 15.0 if looks_like_face_cut(arr, c) else 0.0
        score = gutter_q + balance * 40 - face_pen
        if score > best_score:
            best_score, best_c = score, c
    if best_c is None or best_score < 30:
        return [(y0, y1)]
    left = recursive_split(arr, row_mean, edge_energy, y0, best_c, min_panel, max_depth, depth + 1)
    right = recursive_split(arr, row_mean, edge_energy, best_c, y1, min_panel, max_depth, depth + 1)
    return left + right

def primary_content_blocks(
    row_mean, edge_energy, white_thresh, edge_thresh, min_gutter, min_panel,
    black_thresh: float = 25.0,
) -> List[Tuple[int, int]]:
    """
    Primary gutters = white OR pure-black void rows (manhwa often uses black dividers).
    These are structural separators and take priority over event/projection signals.
    """
    h = len(row_mean)
    white_gutter = (row_mean > white_thresh) & (edge_energy < edge_thresh)
    # pure black void: very dark AND low edge (not detailed dark art)
    black_gutter = (row_mean < black_thresh) & (edge_energy < max(edge_thresh * 1.5, 12.0))
    gutter = white_gutter | black_gutter
    regions = mask_to_regions(gutter, min_len=min_gutter)
    blocks, prev = [], 0
    for g0, g1 in regions:
        if g0 - prev >= min_panel:
            blocks.append((prev, g0))
        prev = g1 + 1
    if h - prev >= min_panel:
        blocks.append((prev, h))
    return blocks if blocks else [(0, h)]

# -----
# YOLO sliding-window path (long strips)
# -----
def yolo_panels_sliding(
    img: Image.Image,
    weights: str,
    conf: float = 0.25,

```

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iou: float = 0.45,
window: int = 1280,
overlap: float = 0.25,
min_panel: int = 80,
) -> List[Tuple[int, int]]:
    """
    Run YOLO on overlapping vertical windows of a tall strip, merge boxes,
    convert to horizontal panel ranges sorted top→bottom.
    """
    if not HAS_YOLO:
        raise RuntimeError("ultralytics not installed. pip install ultralytics")
    model = YOLO(weights)
    arr = np.asarray(img)
    H, W = arr.shape[:2]
    step = max(1, int(window * (1 - overlap)))
    all_boxes = [] # (y1, y2, conf)

    y = 0
    while y < H:
        y_end = min(y + window, H)
        crop = arr[y:y_end]
        # ultralytics accepts numpy RGB
        results = model.predict(crop, conf=conf, iou=iou, verbose=False)
        for r in results:
            if r.bboxes is None:
                continue
            for box in r.bboxes:
                xyxy = box.xyxy[0].cpu().numpy()
                c = float(box.conf[0])
                # map back to full-image coords
                by1 = int(xyxy[1]) + y
                by2 = int(xyxy[3]) + y
                if by2 - by1 >= min_panel:
                    all_boxes.append((by1, by2, c))
            if y_end >= H:
                break
        y += step

    if not all_boxes:
        return []

    # NMS-like merge on 1-D intervals (sort by conf desc)
    all_boxes.sort(key=lambda t: -t[2])
    kept = []
    for y1, y2, c in all_boxes:
        overlap_existing = False
        for ky1, ky2 in kept:
            inter = max(0, min(y2, ky2) - max(y1, ky1))
            union = (y2 - y1) + (ky2 - ky1) - inter
            if union > 0 and inter / union > 0.4:
                overlap_existing = True
                break
        if not overlap_existing:
            kept.append((y1, y2))
    kept.sort(key=lambda t: t[0])

    # Convert possibly overlapping/gapped boxes into contiguous panel ranges
    # by using midpoints between consecutive boxes as cuts when needed
    if not kept:
        return []
    panels = []
    # start from top
    cur_start = max(0, kept[0][0] - 2)
    for i, (y1, y2) in enumerate(kept):
        if i == 0:
            cur_start = max(0, y1 - 2)
            cur_end = y2
        else:
            prev_end = kept[i - 1][1]
            # gap or small overlap → close previous panel at midpoint
            mid = (prev_end + y1) // 2
            panels.append((cur_start, mid))
            cur_start = mid
            cur_end = y2
        cur_end = max(cur_end, y2)
    panels.append((cur_start, min(H, cur_end + 2)))
    # filter tiny
    panels = [(a, b) for a, b in panels if b - a >= min_panel]
    return panels

# -----
# Main classical / hybrid path
# -----
def classical_slice(
    img: Image.Image,
    min_gutter: int = 20,
    min_panel_height: int = 100,
    white_thresh: float = 245.0,
    edge_thresh: float = 8.0,
    subdivide: bool = False,
    subdivide_max_height: int = 2200,
    max_cuts: int = 8,
    mode: str = "hybrid",
    use_hough: bool = False,
) -> List[Tuple[int, int]]:
    arr = np.asarray(img)
    H, W = arr.shape[:2]
    row_mean, row_std, edge_energy = row_statistics(arr)

    blocks = primary_content_blocks(
        row_mean, edge_energy, white_thresh, edge_thresh, min_gutter, min_panel_height
    )

    if subdivide or mode in ("recursive", "hybrid"):
        refined: List[Tuple[int, int]] = []
        for y0, y1 in blocks:
            bh = y1 - y0

```

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        if bh <= subdivide_max_height and mode != "recursive":
            refined.append((y0, y1))
            continue

        if mode == "recursive" or (mode == "hybrid" and bh > subdivide_max_height):
            pieces = recursive_split(
                arr, row_mean, edge_energy, y0, y1,
                min_panel_height, max_depth=max(2, max_cuts // 2),
            )
            refined.extend(pieces)
            continue

        # --- non-overlapping signal priority ---
        # 1) event-zone ends (SFX/impact) – highest for action manhwa
        # 2) projection valleys – structural gutters inside the block
        # 3) Hough lines – only if still sparse
        # Final NMS merges anything closer than min_side (later cut wins)
        min_side = int(min_panel_height * 1.5)
        raw_cuts: List[int] = []

        for _, c in event_zone_cuts(row_mean, y0, y1, min_side):
            raw_cuts.append(c)

        for c in projection_split_candidates(row_mean, edge_energy, y0, y1, min_side):
            raw_cuts.append(c)

        if use_hough and HAS_CV2 and len(raw_cuts) < 2:
            for c in hough_horizontal_candidates(arr, y0, y1, min_side):
                raw_cuts.append(c)

        # face-safety filter
        raw_cuts = [c for c in raw_cuts if not looks_like_face_cut(arr, c)]
        cuts = merge_nearby_cuts(raw_cuts, min_side)[:max_cuts]

        pieces, prev = [], y0
        for c in cuts:
            if c - prev >= min_panel_height // 2:
                pieces.append((prev, c))
                prev = c
            if y1 - prev >= min_panel_height // 2:
                pieces.append((prev, y1))
        elif pieces:
            a, _ = pieces[-1]
            pieces[-1] = (a, y1)
        else:
            pieces.append((y0, y1))
        refined.extend(pieces)
        blocks = refined

    # Merge fragments
    merged: List[Tuple[int, int]] = []
    for y0, y1 in blocks:
        if y1 <= y0:
            continue
        ph = y1 - y0
        if merged and ph < min_panel_height:
            a, _ = merged[-1]
            merged[-1] = (a, y1)
        else:
            merged.append((y0, y1))
    if len(merged) > 1:
        final = [merged[0]]
        for y0, y1 in merged[1:]:
            ph = y1 - y0
            a, b = final[-1]
            prev_h = b - a
            if ph < min_panel_height * 4.0 and ph < prev_h * 0.55:
                final[-1] = (a, y1)
            else:
                final.append((y0, y1))
        merged = final
    return merged

# -----
# Public API
# -----
def slice_panels(
    img: Image.Image,
    min_gutter: int = 20,
    min_panel_height: int = 100,
    white_thresh: float = 245.0,
    edge_thresh: float = 8.0,
    subdivide: bool = False,
    subdivide_max_height: int = 2200,
    max_cuts: int = 8,
    mode: str = "hybrid",
    use_ocr: bool = False,
    use_hough: bool = False,
    yolo_weights: Optional[str] = None,
    yolo_conf: float = 0.25,
    yolo_window: int = 1280,
    pad: int = 4,
) -> List[Panel]:
    arr = np.asarray(img)
    H, W = arr.shape[:2]
    source = "classical"
    blocks: List[Tuple[int, int]] = []

    # Try YOLO first if requested
    if yolo_weights and HAS_YOLO:
        try:
            blocks = yolo_panels_sliding(
                img, yolo_weights, conf=yolo_conf,
                window=yolo_window, min_panel=min_panel_height,
            )
        if blocks:
            source = "yolo"

```

```

except Exception as e:
    print(f"YOLO path failed ({e}); falling back to classical.", file=sys.stderr)
    blocks = []

if not blocks:
    blocks = classical_slice(
        img,
        min_gutter=min_gutter,
        min_panel_height=min_panel_height,
        white_thresh=white_thresh,
        edge_thresh=edge_thresh,
        subdivide=subdivide or mode == "recursive",
        subdivide_max_height=subdivide_max_height,
        max_cuts=max_cuts,
        mode=mode,
        use_hough=use_hough,
    )
    source = "classical"

H, W = arr.shape[:2]
panels: List[Panel] = []
for i, (y0, y1) in enumerate(blocks):
    if y1 <= y0:
        continue
    cy0 = max(0, y0 - pad)
    cy1 = min(H, y1 + pad)
    if cy1 <= cy0:
        cy0, cy1 = y0, y1
    crop = img.crop((0, cy0, W, cy1))
    label = ""
    if use_ocr and HAS_OCR:
        try:
            label = " ".join(pyesseract.image_to_string(crop, timeout=8).split())[120:]
        except Exception:
            label = ""
    panels.append(Panel(
        index=i + 1, y0=cy0, y1=cy1, image=crop,
        label=label, source=source,
    ))
return panels

# -----
# CLI
# -----
def main() -> None:
    p = argparse.ArgumentParser(
        description="Webtoon/manhwa panel slicer v6 - classical + optional YOLO for long strips.",
        formatter_class=argparse.RawDescriptionHelpFormatter,
        epilog=__doc__,
    )
    p.add_argument("input", help="Input PDF or image")
    p.add_argument("-o", "--output", default="panels")
    p.add_argument("--dpi", type=int, default=72)
    p.add_argument("--min-gutter", type=int, default=20)
    p.add_argument("--min-panel", type=int, default=100)
    p.add_argument("--white-thresh", type=float, default=245.0)
    p.add_argument("--edge-thresh", type=float, default=8.0)
    p.add_argument("--subdivide", action="store_true")
    p.add_argument("--subdivide-max", type=int, default=2200)
    p.add_argument("--max-cuts", type=int, default=8)
    p.add_argument("--mode", choices=["hybrid", "recursive", "projection"], default="hybrid")
    p.add_argument("--hough", action="store_true")
    p.add_argument("--ocr", action="store_true")
    p.add_argument("--yolo", type=str, default=None, help="Path to YOLO .pt weights (optional)")
    p.add_argument("--yolo-conf", type=float, default=0.25)
    p.add_argument("--yolo-window", type=int, default=1200, help="Sliding window height for tall strips")
    p.add_argument("--pad", type=int, default=4)
    p.add_argument("--prefix", default="panel")
    p.add_argument("--format", choices=["png", "jpg", "webp"], default="png")
    p.add_argument("--quality", type=int, default=95)
    p.add_argument("-q", "--quiet", action="store_true")
    args = p.parse_args()

    if not os.path.isfile(args.input):
        print(f"Error: file not found: {args.input}", file=sys.stderr)
        sys.exit(1)

    if args.yolo and not HAS_YOLO:
        print("Warning: --yolo given but ultralytics not installed. pip install ultralytics", file=sys.stderr)

    if not args.quiet:
        print(f>Loading {args.input} ...")
    img = load_image(args.input, dpi=args.dpi)
    if not args.quiet:
        print(f>Image size: {img.size[0]}x{img.size[1]}")
        if args.yolo:
            print(f>YOLO weights: {args.yolo}")

    panels = slice_panels(
        img,
        min_gutter=args.min_gutter,
        min_panel_height=args.min_panel,
        white_thresh=args.white_thresh,
        edge_thresh=args.edge_thresh,
        subdivide=args.subdivide or args.mode == "recursive",
        subdivide_max_height=args.subdivide_max,
        max_cuts=args.max_cuts,
        mode=args.mode,
        use_ocr=args.ocr,
        use_hough=args.hough,
        yolo_weights=args.yolo,
        yolo_conf=args.yolo_conf,
        yolo_window=args.yolo_window,
        pad=args.pad,
    )

    os.makedirs(args.output, exist_ok=True)

```



```

if not args.quiet:
    src = panels[0].source if panels else "?"
    print(f"Found {len(panels)} panel(s) via [{src}] → {args.output}/")
for pan in panels:
    name = f"{args.prefix}_{pan.index:02d}.{args.format}"
    out = os.path.join(args.output, name)
    kw = {"quality": args.quality} if args.format in ("jpg", "webp") else {}
    pan.image.save(out, **kw)
    extra = f' "{pan.label[:50]}..." ' if pan.label else ""
    if not args.quiet:
        print(f" {name} ({pan.image.size[0]}×{pan.image.size[1]}){extra}")
if not args.quiet:
    print("Done.")

if __name__ == "__main__":
    main()

```

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